Stochastic Processes — 1060429

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Throughout, a Brownian motion (BM) is standard ($B_0 = 0$) with continuous paths, and $\mathcal{F}_t$ denotes the filtration in use. Apparent typos are pointed out where they occur.

Exercise 4

Stochastic calculus

Throughout, for $f \in L^2$ (adapted, with $\mathbb{E}\int_0^t f^2 ds < \infty$), $\int_0^t f\, dB$ is the L^2-limit of $\int f_n\, dB$ for simple adapted $f_n \to f$ in $L^2(ds \otimes d\mathbb{P})$. For a partition $0 = t_0 < \cdots < t_n = t$ of mesh $|\pi|$ write $\Delta_i = B_{t_i} - B_{t_{i-1}}$ and $\Delta t_i = t_i - t_{i-1}$.

1.

Remark. In (a)ii and (b)i the right-hand side should be $\frac{1}{3}B_t^3 - \int_0^t B_s\, ds$, not $\frac{1}{3}B_t^2 - \dots$. As stated it cannot hold: the left side has mean 0, while $\mathbb{E}[\frac{1}{3}B_t^2 - \int_0^t B_s\, ds] = \frac{t}{3}$. Itô's formula in (b) confirms the corrected form.

(a) i. $f_\pi = \sum t_{i-1}\mathbf{1}_{[t_{i-1},t_i)} \to s$ in L^2, since $\int_0^t |f_\pi(s) - s|^2 ds \le t|\pi|^2$. Summation by parts gives

$$\sum_i t_{i-1}\Delta_i = tB_t - \sum_i B_{t_i}\Delta t_i.$$

By continuity $|\sum_i B_{t_i}\Delta t_i - \int_0^t B_s\, ds| \to 0$ pathwise. It is bounded by $2t\sup_{[0,t]}|B| \in L^2$, so the convergence also holds in L^2. Hence $\int_0^t s\, dB_s = tB_t - \int_0^t B_s\, ds$.

(a) ii. For $u < s \le t$, by Cauchy–Schwarz and $B_s + B_u \sim N(0, s+3u)$, $\mathbb{E}(B_s^2 - B_u^2)^2 \le \sqrt{3(s-u)^2}\sqrt{3(4t)^2} = 12t(s-u)$. Hence $\sum_i B_{t_{i-1}}^2 \mathbf{1}_{[t_{i-1},t_i)} \to B^2$ in $L^2(ds \otimes d\mathbb{P})$, and $\int_0^t B^2 dB = \lim_{L^2} S_\pi$ with $S_\pi = \sum_i B_{t_{i-1}}^2 \Delta_i$. Summing $b^3 - a^3 = 3a^2(b-a) + 3a(b-a)^2 + (b-a)^3$ over the partition gives

$$S_\pi = \tfrac{1}{3}B_t^3 - \sum_i B_{t_{i-1}}\Delta_i^2 - \tfrac{1}{3}\sum_i \Delta_i^3.$$

The cross terms below vanish by independence of increments and since odd moments are 0, resp. $\mathbb{E}[\Delta_j^2 - \Delta t_j] = 0$:

$$\mathbb{E}\Big(\sum_i \Delta_i^3\Big)^2 = \sum_i 15\Delta t_i^3 \le 15t|\pi|^2, \qquad \mathbb{E}\Big(\sum_i B_{t_{i-1}}(\Delta_i^2 - \Delta t_i)\Big)^2 = \sum_i t_{i-1}\cdot 2\Delta t_i^2 \le 2t^2|\pi|.$$

Moreover $\sum_i B_{t_{i-1}}\Delta t_i \to \int_0^t B_s\, ds$ in L^2 as in i. Hence $\int_0^t B_s^2\, dB_s = \frac{1}{3}B_t^3 - \int_0^t B_s\, ds$.

(b) i. Itô's formula with $f(x) = x^3/3$ (so $f' = x^2$, $f'' = 2x$) gives $\frac{1}{3}B_t^3 = \int_0^t B_s^2\, dB_s + \int_0^t B_s\, ds$.

2.

Remark. We take $f, g \in L^2([0,T]\times\Omega)$, i.e. $\mathbb{E}\int_0^T f^2 < \infty$ for every T, the class on which the Itô integral is an L^2-martingale. If only $\int_0^T f^2 ds < \infty$ a.s. were assumed, the claim is false in general; we do not pursue this.

The Itô integral has mean zero, so taking expectations gives $C = D$. Then $\int_0^T (f-g)\, dB = 0$ a.s., and by the Itô isometry

$$\mathbb{E}\int_0^T (f-g)^2 ds = \mathbb{E}\Big(\int_0^T (f-g)\, dB\Big)^2 = 0,$$

i.e. $f = g$ for $ds \otimes d\mathbb{P}$-a.e. (s, ω).

3.

(a) $d(B_t^2) = dt + 2B_t\, dB_t$.

(b) $d(2 + t + e^{B_t}) = (1 + \frac{1}{2}e^{B_t})\, dt + e^{B_t}\, dB_t$.

(c) $d(B_t^2 + \tilde{B}_t^2) = 2\, dt + 2B_t\, dB_t + 2\tilde{B}_t\, d\tilde{B}_t$, i.e. $\alpha = 2$ and $\beta = (2B_t, 2\tilde{B}_t)$ against $d(B_t, \tilde{B}_t)$.

(d) $d(t, B_t) = \binom{1}{0} dt + \binom{0}{1} dB_t$.

(e) With $f(t,x) = e^{t/2}\cos x$: $f_t = \frac{1}{2}f$, $f_x = -e^{t/2}\sin x$, $f_{xx} = -f$. Hence

$$d\big(e^{t/2}\cos B_t\big) = \big(\tfrac{1}{2}f - \tfrac{1}{2}f\big)dt - e^{t/2}\sin B_t\, dB_t = -e^{t/2}\sin B_t\, dB_t.$$

So $e^{t/2}\cos B_t = 1 - \int_0^t e^{s/2}\sin B_s\, dB_s$. The integrand is bounded by $e^{T/2}$ on $[0,T]$, so it lies in L^2 and the Itô integral is a martingale. Therefore $e^{t/2}\cos B_t$ is a martingale.

4.

(a) With $f = e^x$, $f' = f'' = f$: $dX_t = e^{B_t}dB_t + \frac{1}{2}e^{B_t}dt = \frac{1}{2}X_t dt + X_t dB_t$.

(b) With $f = \sin x$: $dX_t = \cos B_t\, dB_t - \frac{1}{2}\sin B_t\, dt$. While $\cos B_t > 0$, i.e. for $t < \inf\{s : |B_s| = \pi/2\}$, we have $\cos B_t = \sqrt{1 - \sin^2 B_t} = \sqrt{1 - X_t^2}$. Hence $dX_t = -\frac{1}{2}X_t dt + \sqrt{1 - X_t^2}\, dB_t$.

5. For $k \geq 2$, Itô's formula gives

$$B_t^k = \int_0^t k B_s^{k-1} dB_s + \tfrac{1}{2}k(k-1)\int_0^t B_s^{k-2} ds.$$

The integrand satisfies $\mathbb{E}\int_0^t B_s^{2k-2} ds < \infty$, so the stochastic integral has mean 0. By Fubini,

$$\beta_k(t) = \tfrac{1}{2}k(k-1)\int_0^t \beta_{k-2}(s)\, ds.$$

The formula also holds for $k = 1$ ($\beta_1 = 0$) but fails for $k = 0$, where $\beta_0 = 1$. With $\beta_0 = 1$: $\beta_2(t) = t$, then $\mathbb{E}B_t^4 = \beta_4(t) = 6\int_0^t s\, ds = 3t^2$ and $\mathbb{E}B_t^6 = \beta_6(t) = 15\int_0^t 3s^2\, ds = 15t^3$.

6.

Remark. g_ε is not C^2 on $\mathbb{R}$. It is C^1, with $g_\varepsilon'(x) = \operatorname{sgn} x$ for $|x| \geq \varepsilon$ and x/ε for $|x| < \varepsilon$. Its second derivative $g_\varepsilon'' = \varepsilon^{-1}\mathbf{1}_{(-\varepsilon,\varepsilon)}$ exists except at $\pm\varepsilon$, so g_ε is C^2 on $\mathbb{R} \setminus \{\pm\varepsilon\}$. Itô's formula still applies, as shown in (a).

(a) Let ϕ_n be continuous, $0 \leq \phi_n \leq 1/\varepsilon$, with $\phi_n = g_\varepsilon''$ outside $(\pm\varepsilon - \frac{1}{n}, \pm\varepsilon + \frac{1}{n})$. Let $h_n(x) = g_\varepsilon(0) + \int_0^x \int_0^y \phi_n$, which is C^2. Since $g_\varepsilon'(0) = 0$, $|h_n' - g_\varepsilon'| \leq \int |\phi_n - g_\varepsilon''| \leq 4/(n\varepsilon)$ and $|h_n - g_\varepsilon|(x) \leq 4|x|/(n\varepsilon)$. Itô's formula for h_n reads

$$h_n(B_t) = h_n(B_0) + \int_0^t h_n'(B_s)\, dB_s + \tfrac{1}{2}\int_0^t \phi_n(B_s)\, ds.$$

As $n \to \infty$: $h_n(B_t) \to g_\varepsilon(B_t)$, and $\int h_n'(B)dB \to \int g_\varepsilon'(B)dB$ in L^2 by the isometry and uniform convergence. Also $\phi_n(B_s) \to g_\varepsilon''(B_s)$ for all s with $B_s \neq \pm\varepsilon$, which is a.e. s a.s. because $\mathbb{E}\,\mathrm{Leb}\{s \leq t : B_s = \pm\varepsilon\} = \int_0^t \mathbb{P}(B_s = \pm\varepsilon)ds = 0$. Bounded convergence then gives

$$g_\varepsilon(B_t) = g_\varepsilon(B_0) + \int_0^t g_\varepsilon'(B_s)\, dB_s + \frac{1}{2\varepsilon}\mathrm{Leb}\{s \in [0,t] : B_s \in (-\varepsilon, \varepsilon)\}.$$

(b) By the isometry and dominated convergence ($\mathbb{P}(B_s = 0) = 0$ for $s > 0$),

$$\mathbb{E}\Big(\int_0^t \tfrac{B_s}{\varepsilon} \mathbf{1}_{\{|B_s|<\varepsilon\}} dB_s \Big)^2 = \mathbb{E} \int_0^t \tfrac{B_s^2}{\varepsilon^2} \mathbf{1}_{\{|B_s|<\varepsilon\}} ds \le \int_0^t \mathbb{P}(|B_s| < \varepsilon)\, ds \xrightarrow[\varepsilon\to 0]{} 0.$$

(c) Split $\int g'_\varepsilon(B) dB = \int \operatorname{sgn}(B) \mathbf{1}_{\{|B|\ge\varepsilon\}} dB + \int \frac{B}{\varepsilon} \mathbf{1}_{\{|B|<\varepsilon\}} dB$. The second term tends to 0 in L^2 by (b). The first tends to $\int_0^t \operatorname{sgn}(B_s) dB_s$ in L^2, since $\mathbb{E}\int_0^t \mathbf{1}_{\{|B_s|<\varepsilon\}} ds \to 0$. Moreover $0 \le g_\varepsilon(x) - |x| = (\varepsilon - |x|)^2/(2\varepsilon)\, \mathbf{1}_{\{|x|<\varepsilon\}} \le \varepsilon/2$. Hence by (a)

$$\frac{1}{2\varepsilon} \mathrm{Leb}\{s \le t : B_s \in (-\varepsilon, \varepsilon)\} = g_\varepsilon(B_t) - g_\varepsilon(B_0) - \int_0^t g'_\varepsilon(B_s) dB_s$$

converges in L^2 to $|B_t| - |B_0| - \int_0^t \operatorname{sgn}(B_s) dB_s$. Thus L_t exists as an L^2-limit and $|B_t| = |B_0| + \int_0^t \operatorname{sgn}(B_s) dB_s + L_t$.

7.

Remark. The definition should read $T_x = \inf\{t > 0 : X_t^x = 0\}$ with $x > 0$, not $\inf\{t : X_t^x > 0\}$. In (b) the claim is $(Y_t) \overset{d}{=} (X_t^1)$. All X^x, $x > 0$, are driven by the same B.

Facts used: X^x is continuous and positive on $[0, T_x)$; $B_t/t \to 0$ a.s.; and the following consequence of the equation. If $X_t^x < R$ for all $t \ge 0$, then $T_x = \infty$ and $X_t^x \ge x + B_t + at/R \to \infty$, a contradiction. So X^x cannot stay in (r, R) forever.

(a) For $t < S = T_x \wedge T_y$, $D_t = X_t^y - X_t^x = y - x + a\int_0^t \big(\frac{1}{X^y} - \frac{1}{X^x}\big) ds$ is C^1 with $D' = -aD/(X^x X^y)$. Hence

$$D_t = (y - x) \exp\Big(- a \int_0^t \frac{ds}{X_s^x X_s^y} \Big) > 0.$$

If $T_y < T_x$, then $0 < X_t^x < X_t^y \to 0$ as $t \uparrow T_y$, so $X_{T_y}^x = 0$, a contradiction. So $T_x \le T_y$ and $X_t^x < X_t^y$ for $t < T_x$. With $y = x$, the same computation gives $D \equiv 0$: the equation has at most one solution for a given path of B (pathwise uniqueness).

(b) Let $W_t = x^{-1} B_{x^2 t}$, a BM. Substituting $s = x^2 u$ in $\int_0^{x^2 t} ds/X_s^x = x^2 \int_0^t du/X_{x^2 u}^x = x \int_0^t du/Y_u$,

$$Y_t = 1 + W_t + a \int_0^t \frac{du}{Y_u}, \qquad t < T_x/x^2.$$

So Y solves the equation with $x = 1$ driven by W, and its lifetime is T_x/x^2. By pathwise uniqueness the solution is a fixed functional of the driving path, $X^1 = \Phi(B)$, and therefore $Y = \Phi(W) \overset{d}{=} \Phi(B) = X^1$.

(c) Let $s(x) = x^{1-2a}$ for $a \ne \frac{1}{2}$ and $s(x) = \log x$ for $a = \frac{1}{2}$. Then $\frac{1}{2}s'' + \frac{a}{x}s' = 0$. For $0 < r < x < R$ let τ be the exit time of X^x from (r, R); it is a.s. finite by the facts above. Itô's formula for $dX = \frac{a}{X} dt + dB$ gives $s(X_{t\wedge\tau}) = s(x) + \int_0^{t\wedge\tau} s'(X) dB$, whose integrand is bounded, so the integral is a martingale. Bounded convergence yields $\mathbb{E}s(X_\tau) = s(x)$, i.e.

$$p_{r,R}(x) := \mathbb{P}(X^x \text{ hits } r \text{ before } R) = \frac{s(R) - s(x)}{s(R) - s(r)}.$$

By (a), $T_x \le T_y$ and $X^x \le X^y$ for $x < y$. Hence it suffices to prove i, ii and iv for the countably many $x \in \mathbb{N}$, and iii for $x = 1/n$, $n \in \mathbb{N}$.

i. ($a \ge \frac{1}{2}$.) If $T_x < \infty$, then $\sup_{[0,T_x]} X^x < R$ for some R, so X^x hits every r before R. But $p_{r,R}(x) \to 0$ as $r \to 0$, since $s(r) \to -\infty$ for $a = \frac{1}{2}$, and for $a > \frac{1}{2}$ we have $s(r) \to +\infty$ while $s(R) - s(x)$ is finite. Hence $\mathbb{P}(T_x < \infty) \le \sum_R \lim_{r\to 0} p_{r,R}(x) = 0$.

ii. ($a = \frac{1}{2}$.) By i, $T_x = \infty$. $\mathbb{P}(X^x \text{ hits } r) \ge p_{r,R}(x) = \frac{\log R - \log x}{\log R - \log r} \to 1$ as $R \to \infty$. Intersecting over $r = 1/n$ gives $\inf_t X_t^x = 0$ a.s.

iii. ($a > \frac{1}{2}$.) Here $T_x = \infty$ and $M_t = (X_t^x)^{1-2a} > 0$. With $\sigma_n = \inf\{t : X_t^x \notin (\frac{1}{n}, n)\} \uparrow \infty$, the process $M_{t\wedge\sigma_n}$ is a bounded martingale. Conditional Fatou gives $\mathbb{E}[M_t|\mathcal{F}_s] \le \liminf_n M_{s\wedge\sigma_n} = M_s$, so M is a non-negative continuous supermartingale. By Exercise 2, Martingales 5(a) applied to $-M$, $M_t \to M_\infty \in [0,\infty)$ a.s., hence $X_t^x \to X_\infty \in (0,\infty]$. If $X_\infty < \infty$ then X^x is bounded, which contradicts the facts above. So $X_t^x \to \infty$.

iv. ($a < \frac{1}{2}$.) Here $s(0) = 0$ and $s(\infty) = \infty$. The events $E_r = \{X^x \text{ hits } r \text{ before } R\}$ decrease as $r \downarrow 0$. On $\bigcap_r E_r$ the hitting times τ_r increase to some $\sigma \le \tau_R$. Moreover $\sigma < \infty$ a.s.: $\tau_R < \infty$ when $T_x = \infty$ by the facts above, and $\sigma \le T_x$ otherwise. By continuity $X_\sigma^x = 0$, so $\sigma = T_x < \tau_R$. Hence

$$\mathbb{P}(T_x < \tau_R) = \lim_{r\to 0} p_{r,R}(x) = 1 - (x/R)^{1-2a} \to 1 \quad (R \to \infty),$$

i.e. $T_x < \infty$ a.s.

For v and vi fix $y > x > 0$, write $X = X^x$, $D = X^y - X^x$, and let $V = X/D$ on $[0, T_x)$. D is C^1, $D > 0$ on $[0, T_x)$, and $T_x < \infty$ a.s. by iv. By (a), $T_x = T_y$ iff $D_{T_x} = 0$. From $d(1/D) = \frac{a}{XX^yD}dt$, $X = DV$ and $X^y = D(V+1)$,

$$dV = \frac{1}{D}\Big(\frac{a}{X}dt + dB\Big) + X\,d\Big(\frac{1}{D}\Big) = \frac{1}{D^2}\Big(\frac{a}{V} + \frac{a}{V+1}\Big)dt + \frac{1}{D}\,dB.$$

For $\varphi \in C^2(0,\infty)$, Itô's formula gives $d\varphi(V) = D^{-2}\big(\frac{1}{2}\varphi'' + (\frac{a}{V} + \frac{a}{V+1})\varphi'\big)(V)\,dt + \varphi'(V)D^{-1}dB$. Let

$$S(v) = \int_0^v \big(u(1+u)\big)^{-2a}du,$$

which is finite since $a < \frac{1}{2}$. Then $\frac{1}{2}S'' + (\frac{a}{v} + \frac{a}{v+1})S' = 0$, and $S(V_t)$ has no dt term. Let $\rho_n = \inf\{t < T_x : D_t \le \frac{1}{n} \text{ or } V_t \le \frac{1}{n}\} \wedge n \wedge T_x$. The integrand of $S(V)$ is bounded up to ρ_n, so $\mathbb{E}S(V_{\rho_n\wedge\kappa}) = S(V_0)$ for every stopping time κ. Moreover $\rho_n \uparrow T_x$: if the limit were $< T_x$, then D and V would be bounded away from 0 on a compact interval past it.

v. ($\frac{1}{4} < a < \frac{1}{2}$.) Now $S(\infty) < \infty$, since $(u(1+u))^{-2a} \sim u^{-4a}$ and $4a > 1$. Suppose $\mathbb{P}(T_x < T_y) = 1$. On $\{T_x < T_y\}$ we have $D_{T_x} > 0$ and $X_t \to 0$, so $V_t \to 0$ as $t \uparrow T_x$. Hence for large n, ρ_n is the hitting time of $\frac{1}{n}$ by V and $S(V_{\rho_n}) = S(\frac{1}{n}) \to 0$. Since $0 \le S \le S(\infty)$, bounded convergence gives $S(V_0) = \lim_n \mathbb{E}S(V_{\rho_n}) = 0$, contradicting $V_0 = x/(y-x) > 0$. Hence $\mathbb{P}(T_x = T_y) > 0$.

vi. ($a \le \frac{1}{4}$.) Now $S(\infty) = \infty$. Let $\kappa_\lambda = \inf\{t : S(V_t) \ge \lambda\}$. Since $S \ge 0$, $\lambda\mathbb{P}(\kappa_\lambda \le \rho_n) \le \mathbb{E}S(V_{\rho_n\wedge\kappa_\lambda}) = S(V_0)$. Letting $n \to \infty$ and then $\lambda \to \infty$ gives $\sup_{t<T_x} V_t < \infty$ a.s. Now fix K and let $\rho_n' = \inf\{t : D_t \le \frac{1}{n} \text{ or } V_t \ge K\} \wedge n \wedge T_x$. Taking expectations in the equation for V (with $d\sigma = dt/D^2$, and the stochastic integral having bounded integrand $1/D \le n$),

$$\mathbb{E}\int_0^{\rho_n'} \frac{a}{V_t}\frac{dt}{D_t^2} \le \mathbb{E}V_{\rho_n'} \le K.$$

On $\{\sup_{t<T_x} V < K\}$ we have $\rho_n' \uparrow T_x$, so monotone convergence gives $\int_0^{T_x} \frac{dt}{V_tD_t^2} < \infty$ a.s. there. Since $\frac{1}{XX^y} = \frac{1}{D^2V(V+1)} \le \frac{1}{D^2V}$,

$$\log D_{T_x} = \log(y-x) - a\int_0^{T_x} \frac{dt}{X_tX_t^y} > -\infty,$$

i.e. $D_{T_x} > 0$ and $T_x < T_y$. Letting $K \to \infty$ along $\mathbb{N}$, $T_x < T_y$ a.s.